Department of Mathematics and Applied Mathematics

3MS Metric Spaces

Tutorial 4 **2024**

1. Let $W = \{(x, y) \in \mathbb{R}^2 \,:\, x > 0\}$.
   Let $A = \{(x, y) \in W \,:\, d_E((1, 0), (x, y)) \leq 2\}$.

   (a) Sketch A.

   (b) Is A a closed subset of $(\mathbb{R}^2, d_E)$?

   (c) Is A a closed subset of (W, d_E)?

   (d) Is A a closed subset of (A, d_E)?

2. Let $A = \{a + bi \in \mathbb{C} \,:\, a > 4\}$ and $B = \{a + bi \in \mathbb{C} \,:\, a \geq 4\}$.
   Let δ be defined by

$$\delta(z, w) = \begin{cases} 0 & \text{if } z = w \\ |z| + |w| & \text{if } z \neq w. \end{cases}$$

   (a) In $(\mathbb{C}, d_E)$ is A open? is B open?

   (b) In $(\mathbb{C}, \delta)$ is A open? is B open?

3. **Definition** Let (X, d) be a metric space and $A \subseteq X$. We say that A is a **dense** subset of X (or, A is dense in X) if $\overline{A} = X$.
   For example, using the usual metric on the reals: (1) $(0, 1)$ is dense in $[0, 1]$, but $(0, 1)$ is not dense in $\mathbb{R}$ (2) $\mathbb{Q}$ is dense in $\mathbb{R}$ and $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$.

   Prove that A is dense in X if and only if A meets every open ball of X.

   (The word "meets" means "has non-empty intersection with"). This is an easy consequence of Proposition 4.17 in the notes which was also proved in class. I recommend that you try and prove it directly (by which I mean use the definition of closure) and only look at Proposition 4.17 if you get stuck.

4. Let C be the Cantor set.

   (a) Is $\dfrac{1}{9} \in C$? Is $\dfrac{1}{9}$ in the boundary of C?

   (b) Give a very quick reason explaining why you know that $\text{bd}(C) \subseteq C$.

   (c) Show that $\text{bd}(C) = C$.

5. Read the following incorrect argument, and then answer the questions below.

False Statement Let (X, d) be a metric space, and $A, B \subseteq X$.
Then $\mathrm{bd(A \cup B)} = \mathrm{bd(A)} \cup \mathrm{bd(B)}$.
Incorrect Argument Let (X, d) be a metric space, and $A, B \subseteq X$. (1)
Then $\overline{A} = A \cup \mathrm{bd(A)}$. (2)
Also $\overline{B} = B \cup \mathrm{bd(B)}$. (3)
Also $\overline{A \cup B} = (A \cup B) \cup \mathrm{bd(A \cup B)}$. (4)
But $\overline{A} \cup \overline{B} = \overline{A \cup B}$. (5)
So $(A \cup B) \cup \mathrm{bd(A)} \cup \mathrm{bd(B)} = \mathrm{(A \cup B)} \cup \mathrm{bd(A \cup B)}$. (6)
So $\mathrm{bd(A)} \cup \mathrm{bd(B)} = \mathrm{bd(A \cup B)}$. (7)

(a) Where does this argument go wrong?

(b) Provide a counter-example to show the statement false.

(c) While you're at it, can you find examples of non-empty sets A, B in a metric space for which $\mathrm{bd(A \cup B)} = \mathrm{bd(A)} \cup \mathrm{bd(B)}$?